

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“SDET - Technology Team”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

ASHIF ALAM - 1DS20AI012

Conducted at



ZopSmart

Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78

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CERTIFICATE

This is to certify that the Internship titled “**SDET - Technology Team**” carried out by **Mr ASHIF ALAM [1DS20AI012]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (20AI8ICINT)

Signature of Reporting Head
Nitin Gupta
Engineering Manager
ZopSmart

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **ASHIF ALAM**, final year student of Information Science and Engineering, Dayananda Sagar College Of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, in **ZopSmart**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:22/04/2024

Place: Bengaluru

USN : 1DS20AI012

Name : Ashif Alam

OFFER LETTER



21st December 2023

Ashif Alam
Bangalore

OFFER LETTER

Dear Ashif,

This has reference to your interest in taking up a full-time employment with **ZopSmart (Raramuri Consultancy Services LLP)**.

We are pleased to offer you a full time employment in the position of **Software Development Engineer in Test – Technology Team**. You will be on internship for a period of 6 months from the date of your joining on **5th February 2024** with a stipend of **Rs. 30,000** per month. And once the internship period is over, you will be converted to full time employee. You will be reporting to the **Manager**, or any other person authorized by him.

1. Position

1.1. You will be employed with the **Technology Team**. In addition to your usual duties, you will also perform, observe, and conform to such directions and instructions assigned or communicated to you by the Company. As a member of an organization that practices flexibility and continuous improvement in work processes and practices, the Company may from time to time change your duties and responsibilities at its sole discretion.

2. Location

2.1 You will be based at the Company's office at **Bangalore**.

2.2 The Company reserves the right to transfer you at any time to any other location in India or abroad, and/or to a branch office, subsidiary, affiliate company, department, or establishment of the Company, whether in existence on the date of this letter or established or acquired later, provided that the terms and conditions of your employment set forth herein are not adversely affected by such transfer. The Company further reserves the right, subject to applicable laws, to transfer your employment to any other company or legal entity, as part of any transfer of relevant business of the Company or as part of any restructuring or amalgamation or reorganization plan implemented by the Company or by which the Company is bound

3. Compensation and Benefits

3.1. Your gross compensation shall be as set out in **Annexure A** annexed hereto and will be subject to all statutory deductions that the Company is required to make.

3.2. The Company intends to provide all the applicable statutory benefits and such other benefits that the Company may from time to time decide to provide including (a) Employee Provident Fund, (b) Employee State Insurance (ESI), (c) Pension. It is however clarified that such benefits and other benefits, if any, and entitlements that may be provided to you by the Company are provided to you on a voluntary basis by the Company in accordance with applicable Company policy and henceare

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr. Nitin Gupta** , ZopSmart, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Coordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of the AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

ASHIF ALAM
[1DS20AI012]

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CHAPTER 1

INTRODUCTION

During my time at Zopsmart, I was immersed in various projects that allowed me to contribute meaningfully to the company's objectives. From collaborating with seasoned developers to testing new features and functionalities, I gained firsthand experience in the entire software development lifecycle.

One of the highlights of my internship was being involved in the development of a new user interface feature aimed at enhancing the navigation experience for customers. This involved not only coding and implementation but also conducting user acceptance testing to ensure the feature met the needs and expectations of our target audience.

Additionally, I had the opportunity to work on optimizing the performance of the e-commerce platform, conducting thorough analysis and implementing solutions to improve loading times and overall responsiveness. This experience honed my problem-solving skills and deepened my understanding of software optimization techniques.

Furthermore, I actively participated in team meetings and brainstorming sessions, where I was encouraged to share ideas and contribute to strategic discussions. This exposure to the decision-making process within a tech company was invaluable and provided me with insights into the broader aspects of product development and management.

Overall, my internship at Zopsmart was not just about acquiring technical skills but also about learning to work effectively within a team, communicate ideas clearly, and adapt to the fast-paced nature of the tech industry. I am grateful for the opportunity to have been part of such an innovative and forward-thinking company, and I am confident that the experiences and lessons learned during my internship will serve me well in my future endeavors.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Zopsmart is dedicated to supporting offline businesses in their transition to the digital world by providing the necessary tools and techniques to establish their online presence. Our overarching goal is to deliver comprehensive solutions that add value for digitally savvy customers. We collaborate closely with businesses to demonstrate how they can enhance customer engagement and innovate within predictable budgets by taking their products online quickly. Through our human-centered design approach, we help businesses reimagine their business models and consumer interactions, leveraging proven analytics to accelerate time-to-market.

Our focus is on enabling businesses to establish their online stores efficiently, reaching new customers and expanding their market reach. By creating end-to-end solutions tailored to their specific needs and goals, we ensure that our customers realize tangible value from their digital endeavors. We emphasize the importance of fostering strong customer relationships and brand loyalty through effective online engagement.

Moreover, our streamlined processes enable businesses to innovate within budget constraints, maximizing their return on investment. We help businesses launch their online stores rapidly, minimizing delays and capitalizing on market opportunities. Through human-centered design principles, we ensure that our solutions are intuitive and user-friendly, meeting the needs of both businesses and their customers.

By leveraging proven analytics, we provide valuable insights into customer behavior, market trends, and performance metrics, empowering businesses to make informed, data-driven decisions. Overall, Zopsmart is committed to empowering businesses to thrive in the digital age by guiding them through the process of establishing and optimizing their online presence, ultimately driving growth and success.

CHAPTER 3

SCOPE OF WORK

At Zopsmart, our scope of work is centered around facilitating the digital transformation of offline businesses and optimizing their online presence. We begin by collaborating closely with each client to develop a tailored digital transformation strategy that aligns with their unique goals and objectives. This strategy encompasses various aspects, including the setup of online stores, website development, design, and integration of payment gateways to ensure a seamless shopping experience.

Our focus extends beyond mere setup; we assist businesses in enhancing customer engagement through strategies like social media marketing, email campaigns, and personalized content creation. We also emphasize innovation within predictable budgets, helping clients maximize their return on investment while ensuring a swift time-to-market through streamlined processes and agile methodologies.

Central to our approach is human-centered design, ensuring that all solutions are intuitive, user-friendly, and align with both business and customer needs. We provide businesses with valuable insights into customer behavior, market trends, and performance metrics through advanced analytics, enabling data-driven decision-making and optimization of their online presence. In essence, our scope of work at Zopsmart encompasses guiding businesses through every step of their digital journey, from initial strategy development to ongoing optimization, driving growth and success in the digital age.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

At Zopsmart, our learning objectives are carefully crafted to empower individuals with the essential skills needed to thrive in the rapidly evolving digital landscape. Our comprehensive training program covers key areas such as API testing, UI testing, mobile UI testing, and CI/CD pipeline implementation, ensuring participants are well-equipped to tackle various challenges in digital development.

Through our API testing module, participants will delve into the intricacies of RestAssured, mastering techniques for seamless testing and validation of RESTful APIs. They will learn to automate API requests, verify responses, and ensure the reliability and accuracy of API testing processes, critical for ensuring the robustness of digital solutions.

In the UI testing segment, participants will harness the power of Selenium to automate tests for web interfaces, enabling efficient validation of user interactions and navigation flows. They will gain insights into cross-browser testing and learn best practices for ensuring the functionality and usability of web applications across different platforms.

Mobile UI testing is another crucial aspect of our training program, where participants will focus on Espresso for Android applications like Zooping. They will learn to automate user interactions and validate UI components, ensuring a seamless and intuitive user experience on mobile devices.

Moreover, participants will gain hands-on experience in implementing CI/CD pipelines using GitHub Actions. They will learn how to set up automated workflows for continuous integration and deployment, streamlining the software delivery process for projects such as EazyUpdates. This knowledge is invaluable for ensuring efficient development cycles and delivering high-quality software products to market swiftly.

By mastering these skills, participants will be well-prepared to drive digital transformation initiative and deliver mature, end-to-end digital solutions for businesses. Our training program reflects Zopsmart's commitment to providing comprehensive and practical learning experiences that empower individuals to excel in the digital age.

CHAPTER 5

CHALLENGES AND SOLUTIONS

In the realm of digital development, several challenges often arise, and at Zopsmart, we provide practical solutions to overcome them. One common challenge is ensuring the reliability and accuracy of API testing. Many developers find it challenging to automate API requests, verify responses, and maintain consistency across different environments. To address this, our training program focuses on mastering tools like RestAssured, enabling participants to automate testing processes effectively. We emphasize best practices for structuring tests, handling authentication, and validating responses, ensuring reliable and accurate API testing.

Another challenge lies in UI testing, particularly the complexity of ensuring consistent functionality across various browsers and devices. Selenium offers robust capabilities for automating web interface tests, but configuring tests for different browsers and platforms can be daunting. Our solution involves comprehensive training on Selenium, covering cross-browser testing techniques and best practices for handling dynamic web elements. Participants learn to create resilient tests that validate user interactions and ensure consistent functionality across different environments.

Mobile UI testing presents its own set of challenges, including the diversity of Android devices and the complexity of validating UI components. With Espresso, participants learn to automate user interactions and validate UI components within Android applications like Zooping. Our training program provides hands-on experience in handling device fragmentation, implementing UI tests for various screen sizes and resolutions, and ensuring a seamless user experience across different devices.

Implementing CI/CD pipelines introduces challenges related to setting up automated workflows and ensuring the reliability of the deployment process. GitHub Actions offers a powerful solution for automating CI/CD pipelines, but configuring workflows and managing dependencies can be complex. Our training program guides participants through the process of setting up automated workflows for continuous integration and deployment in projects like EazyUpdates. They learn to manage build dependencies, run tests automatically, and deploy changes reliably, ensuring efficient software delivery.

Overall, our training program at Zopsmart addresses these challenges by providing practical solutions and hands-on experience with industry-standard tools. Participants graduate with the skills and confidence needed to tackle real-world development challenges and drive digital transformation initiatives effectively.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

I successfully achieved proficiency in various key technologies including Espresso for mobile UI testing, Rest Assured for API testing, Selenium for UI testing, and GitHub Actions for CI/CD pipeline automation. Through dedicated learning and hands-on practice, I mastered the intricacies of each tool, enabling me to contribute effectively to both the EazyUpdates project and the Zooping e-commerce platform.

In the realm of mobile UI testing, I became proficient in Espresso, utilizing it to automate user interactions and validate UI components within Android applications. This achievement facilitated efficient testing of mobile interfaces, ensuring a seamless user experience for applications like Zooping. Additionally, I acquired expertise in API testing with Rest Assured, enabling seamless testing and validation of RESTful APIs. This skill allowed me to automate API requests, verify responses, and ensure the reliability and accuracy of API testing processes.

Furthermore, I honed my skills in UI testing using Selenium, automating tests for web interfaces and user interactions. This proficiency enabled me to validate user interfaces, navigation flows, and functionality across different browsers, enhancing the overall quality of web applications within the EazyUpdates project and beyond.

In terms of project contributions, I played a pivotal role in implementing GitHub Actions for CI/CD pipeline automation for the EazyUpdates project. This automated workflow ensured efficient and reliable software delivery, streamlining the development process and enhancing project management capabilities.

Moreover, my contributions to the Zooping e-commerce platform included the implementation of features such as real-time delivery tracking and promotional offers. These enhancements aimed to improve customer satisfaction and engagement, ultimately contributing to the platform's success in providing a convenient and hassle-free shopping experience.

CHAPTER 7

REFLECTION AND ANALYSIS

Reflecting on my journey at Zopsmart and the skills acquired in mobile UI testing with Espresso, API testing with Rest Assured, UI testing with Selenium, and CI/CD pipeline automation with GitHub Actions, I recognize the transformative impact of continuous learning and application of new technologies.

One significant aspect of my growth has been the understanding of the interconnectedness of different layers in software development. Mastering Espresso for mobile UI testing has not only allowed me to ensure the quality of user experiences but also deepened my appreciation for the intricacies of frontend development. This experience has emphasized the importance of attention to detail and user-centric design principles in crafting digital solutions.

Similarly, delving into API testing with Rest Assured has provided insights into the backend infrastructure that powers modern applications. Understanding how to automate API requests and validate responses has enabled me to ensure the reliability and functionality of crucial backend services, contributing to the overall robustness of digital products.

Moreover, my exploration of UI testing with Selenium has broadened my understanding of web development, particularly in the context of cross-browser compatibility and responsive design. By automating tests for web interfaces, I have gained valuable insights into the challenges of ensuring consistent functionality across diverse platforms and devices.

Implementing GitHub Actions for CI/CD pipeline automation for the EazyUpdates project has been a significant learning experience. It has taught me the importance of automation in streamlining development workflows and ensuring the timely delivery of high-quality software. This experience has underscored the value of continuous integration and deployment practices in maintaining project momentum

and facilitating collaboration among team members.

Furthermore, my contributions to the Zooping e-commerce platforms have reinforced the importance of understanding user needs and preferences in digital product development. Implementing features such as real-time delivery tracking and promotional offers has required a combination of technical expertise and user empathy, highlighting the critical role of customer-centric design in driving engagement and satisfaction.

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CHAPTER 8

RECOMMENDATIONS

Continuous Improvement: Encourage a culture of continuous improvement within the organization, fostering innovation, and embracing new technologies to enhance service quality and efficiency further.

Staff Training and Development: Invest in ongoing training and development programs to ensure that staff members are equipped with the necessary skills and knowledge to effectively utilize new technologies and tools in their roles.

Adopt Frameworks for Testing: Utilize RestAssured for API testing, as it proved instrumental in testing APIs related to critical functionalities such as daily work updates and product catalog management.

Implement Selenium for web UI testing to validate user interfaces, navigation flows, and functionality across various browsers, similar to its use in EazyUpdates.

Integrate Espresso for mobile UI testing, especially for Android applications like Zooping, to ensure a seamless shopping experience on different devices.

Leverage TestNG for Test Management: Implement TestNG for efficient test management and reporting, allowing for organized test execution and detailed analysis of results. Ensure thorough test coverage and maintain test suites for ongoing development and maintenance, contributing to the overall quality of the software solutions.

Automate CI/CD Pipelines with GitHub Actions: Streamline testing and deployment processes by leveraging GitHub Actions for continuous integration and delivery. Automate code integration, testing, and deployment to ensure code integrity and efficient delivery of updates to production environments, similar to its use in both projects.

CHAPTER 9

CONCLUSION

Working on the EazyUpdates and Zooping projects has demonstrated the effectiveness of leveraging frameworks like RestAssured, Selenium, and Espresso, along with tools such as TestNG and GitHub Actions. These frameworks played a pivotal role in ensuring comprehensive testing and validation of critical functionalities within both projects.

For EazyUpdates, RestAssured was instrumental in testing APIs related to daily work updates, organizational features, and project timelines. It enabled seamless automation of API requests and verification of responses, ensuring the reliability and accuracy of backend services. Similarly, in Zooping, RestAssured facilitated testing of the extensive product catalog, order management functionalities, and payment integrations, ensuring a smooth user experience.

Selenium and Espresso were essential for UI testing across both projects. Selenium automated testing of web interfaces for EazyUpdates, validating user interfaces, navigation flows, and functionality across various browsers. Meanwhile, Espresso ensured mobile UI testing for Zooping, guaranteeing a seamless shopping experience on different Android devices.

TestNG provided efficient test management and reporting capabilities, allowing for organized test execution and detailed analysis of results. This facilitated quick issue identification and resolution, contributing to the overall quality of the software solutions.

GitHub Actions streamlined the testing and deployment processes for both projects. By automating CI/CD pipelines, it ensured code integrity, facilitated continuous integration, and enabled efficient delivery of updates to production environments. This automation not only improved development workflows but also enhanced the overall success and reliability of the software solutions.

In conclusion, the successful application of these frameworks and tools underscores their importance in ensuring quality, reliability, and efficiency in software development. Leveraging these technologies has allowed us to deliver robust solutions that meet the needs of our clients and users while maintaining high standards of performance and usability.

CHAPTER 10

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